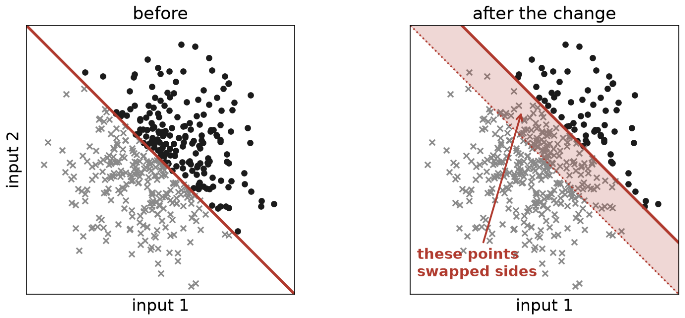
The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

Alexandre Boyer   Mélaine Gouillou   Salomé Fonvielle   Ulysse Petit-Tichanné

Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

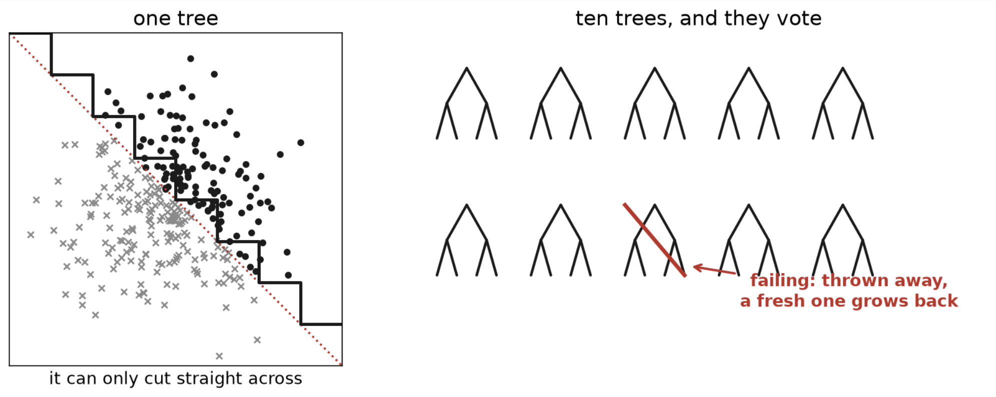
Concept drift: the rule changes, the data does not



one dot = one observation; its shape is the answer the model has to predict

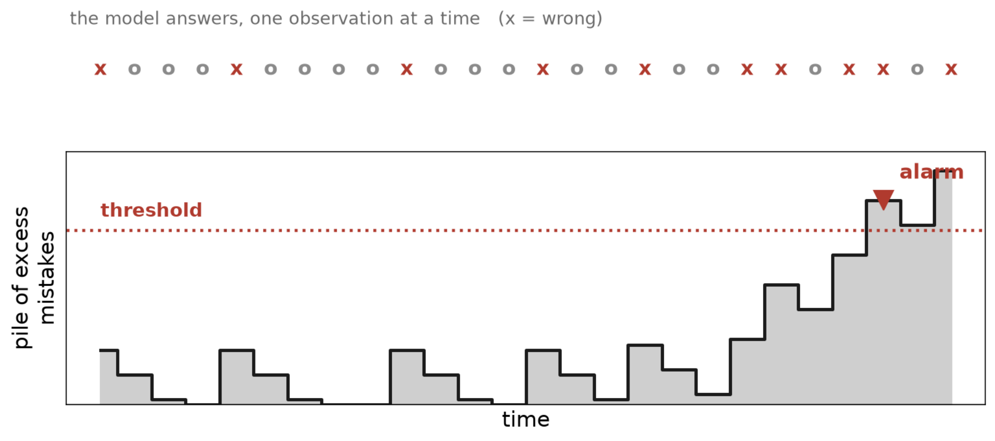
A model trained on the left keeps predicting, confidently, and is now wrong on the shaded band.

Defence one: a model that repairs itself



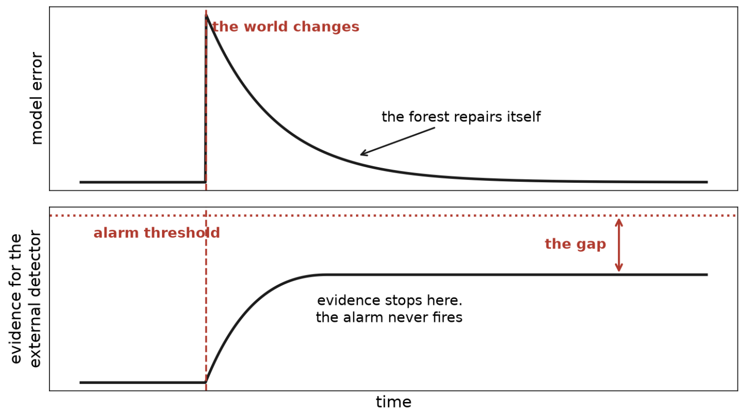
This is the Adaptive Random Forest. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm



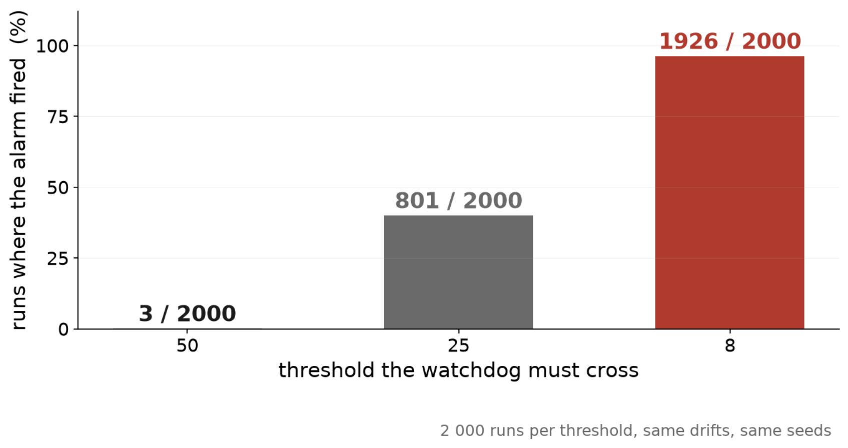
Two communities, two tools, never measured against each other.

Put them together, and they cancel out



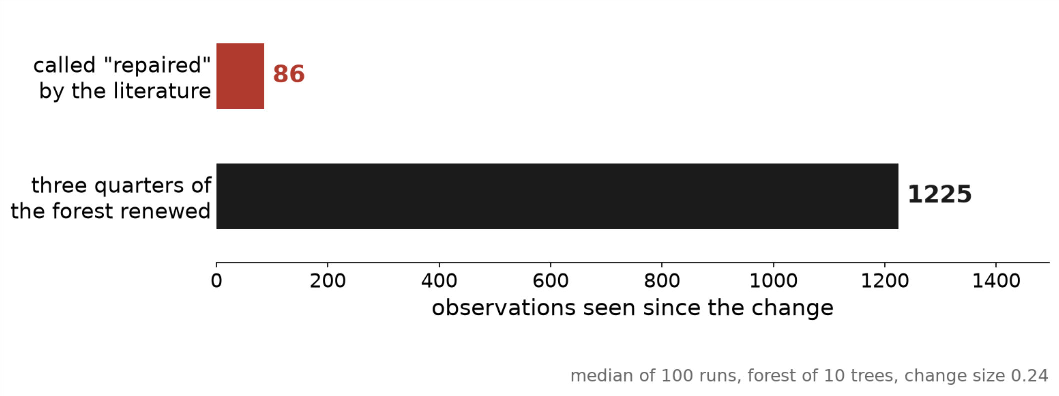
schematic

Why not just lower the threshold?



A low threshold also fires when nothing has happened, and an alarm that cries wolf gets switched off.

Our contribution: the accepted clock is wrong



It marks the first tree out of ten, not a repaired forest.

Where we stand

- The paradox is real and reproducible, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is biased, always the same way. Every conclusion drawn from it inherits that bias.
- **Next**: watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.